

BPL_TEST2_Batch - demo

```
In [1]: # Import the application model and context variables
        from BPL_TEST2_Batch_pyfmi import*
```

Windows - run FMU pre-compiled JModelica 2.14

```
In [2]: # Import the general FMU_explore functions ana adapt them to the application
        from fmu_explore_pyfmi import AdaptWith

        Adapted = AdaptWith(parValue, parLocation, parCheck, \
                               model, fmu_model, fmu_process_diagram, \
                               MSL_usage, MSL_version, BPL_version, \
                               opts_std, simulationTime, timeDiscreteStates, \
                               diagrams, ax, lines)

        par = Adapted.par
        init = Adapted.init
        disp = Adapted.disp
        simu = Adapted.simu
        show = Adapted.show

        resetPen = Adapted.resetPen
        process_diagram = Adapted.process_diagram
        describe_general = Adapted.describe_general
        describe_parts = Adapted.describe_parts
        describe_MSL = Adapted.describe_MSL
        FMU_explore_info = Adapted.FMU_explore_info
        system_info = Adapted.system_info
        SDG = Adapted.SDG
```

```
In [3]: # Bring in the user made fmu_explore functions newplot() and describe()
```

```
In [4]: run -i BPL_TEST2_Batch_explore.py
```

Model for the process has been setup. Key commands:

- par() - change of parameters and initial values
- init() - change initial values only
- simu() - simulate and plot
- newplot() - make a new plot
- show() - show plot from previous simulation
- disp() - display parameters and initial values from the last simulation
- describe() - describe culture, broth, parameters, variables with values/units

Note that both disp() and describe() takes values from the last simulation and the command process_diagram() brings up the main configuration

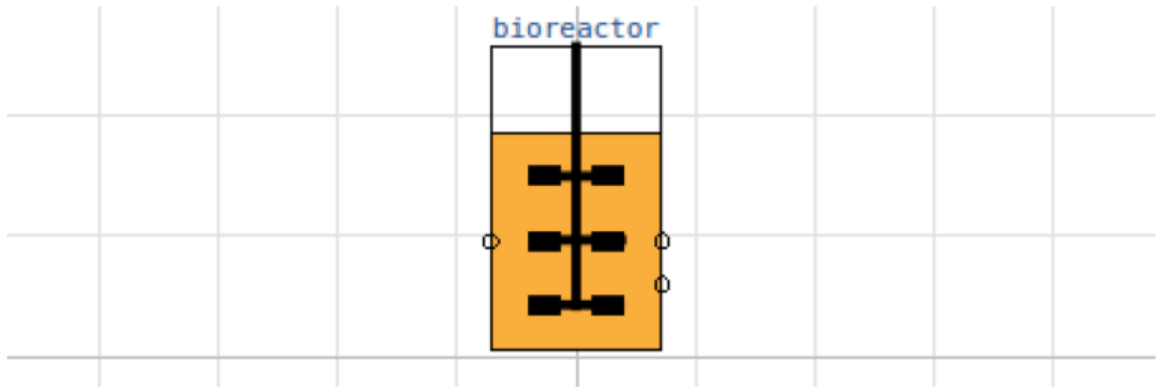
Brief information about a command by help(), eg help(simu)

Key system information is listed with the command system_info()

```
In [5]: %matplotlib inline
        plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]
```

```
In [6]: process_diagram()
```

No processDiagram.png file in the FMU, but try the file on disk.



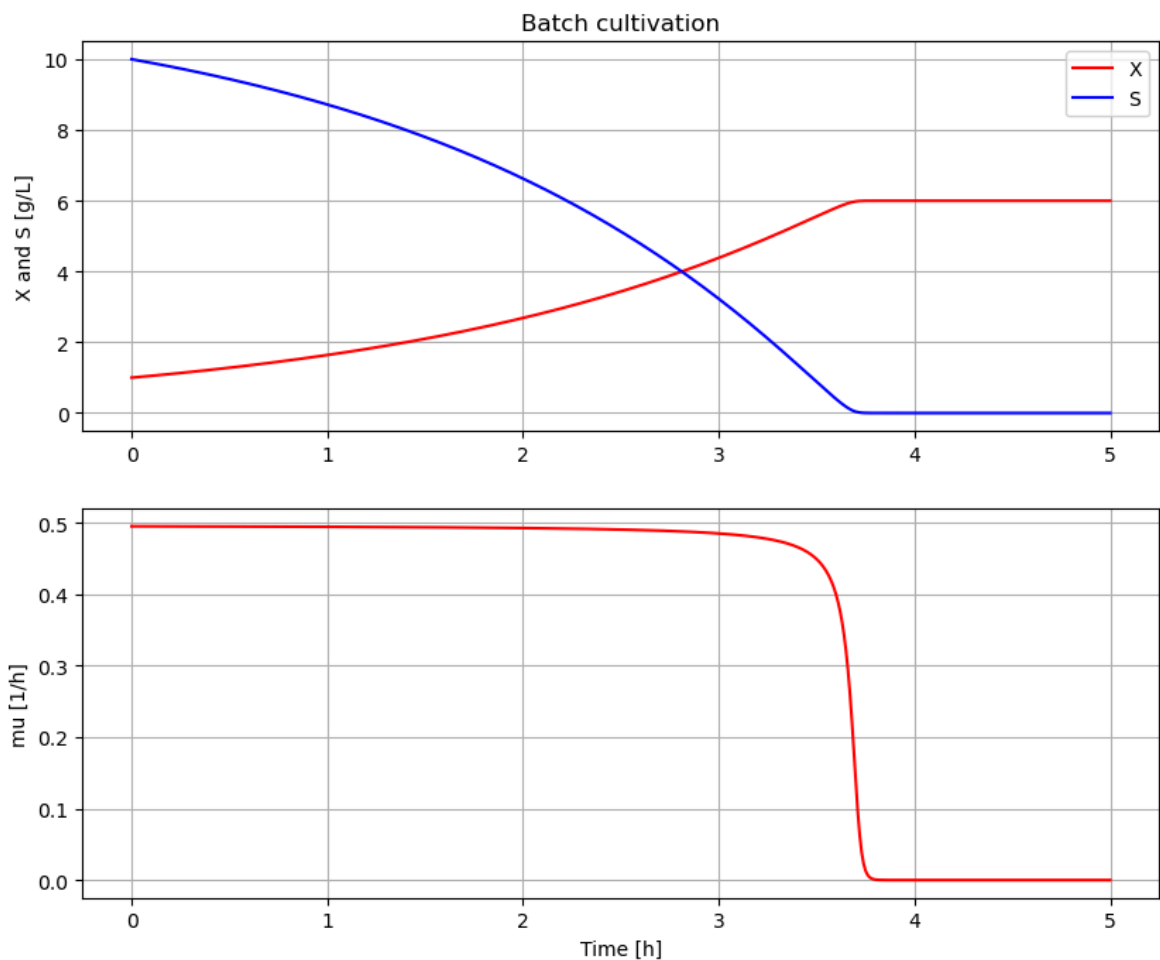
```
In [7]: describe('process')
```

```
In [8]: describe('culture')
```

Simplified text book model - only substrate S and cell concentration X

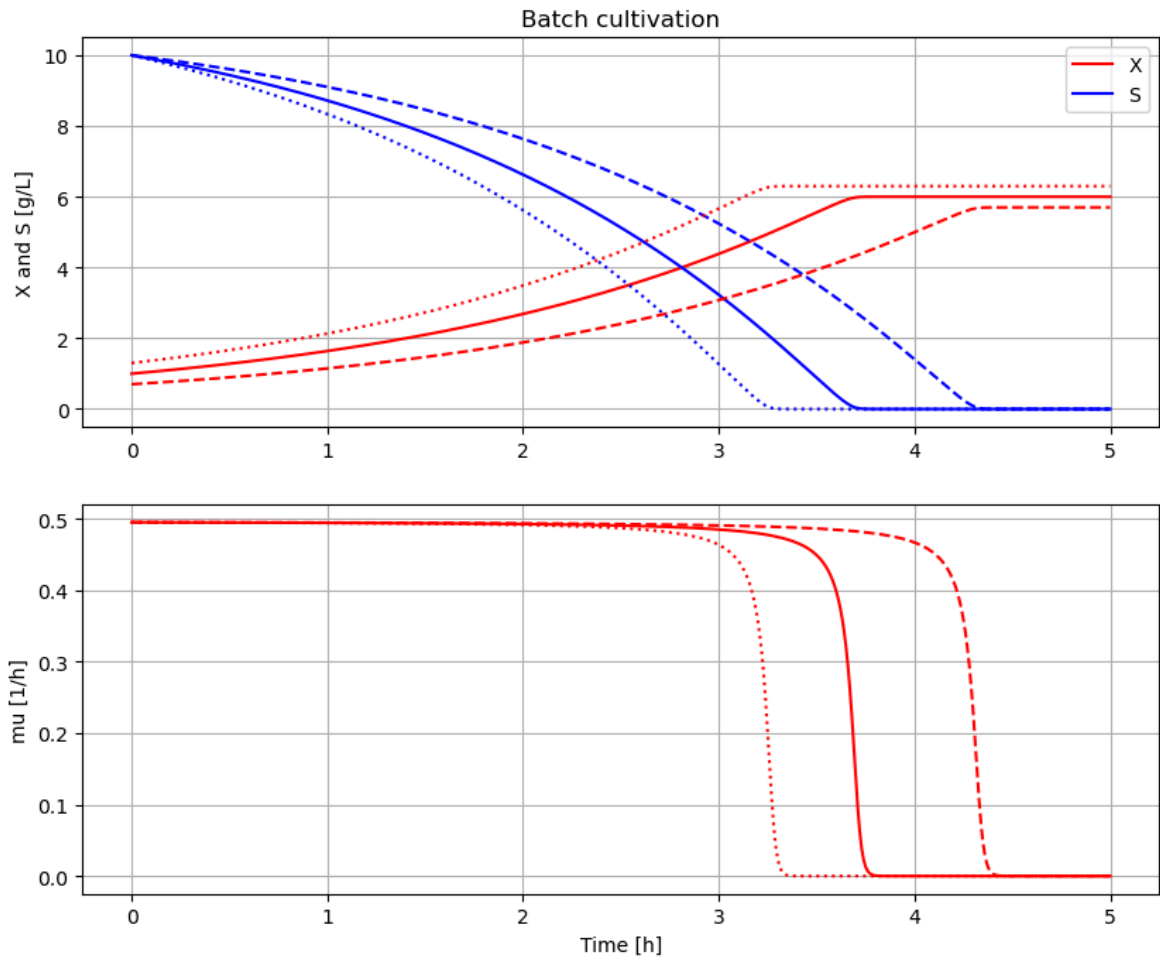
```
disp('bioreactor', mode='long')
```

```
In [9]: # Simulation with default values of the process
newplot(plotType='TimeSeries')
simu()
```



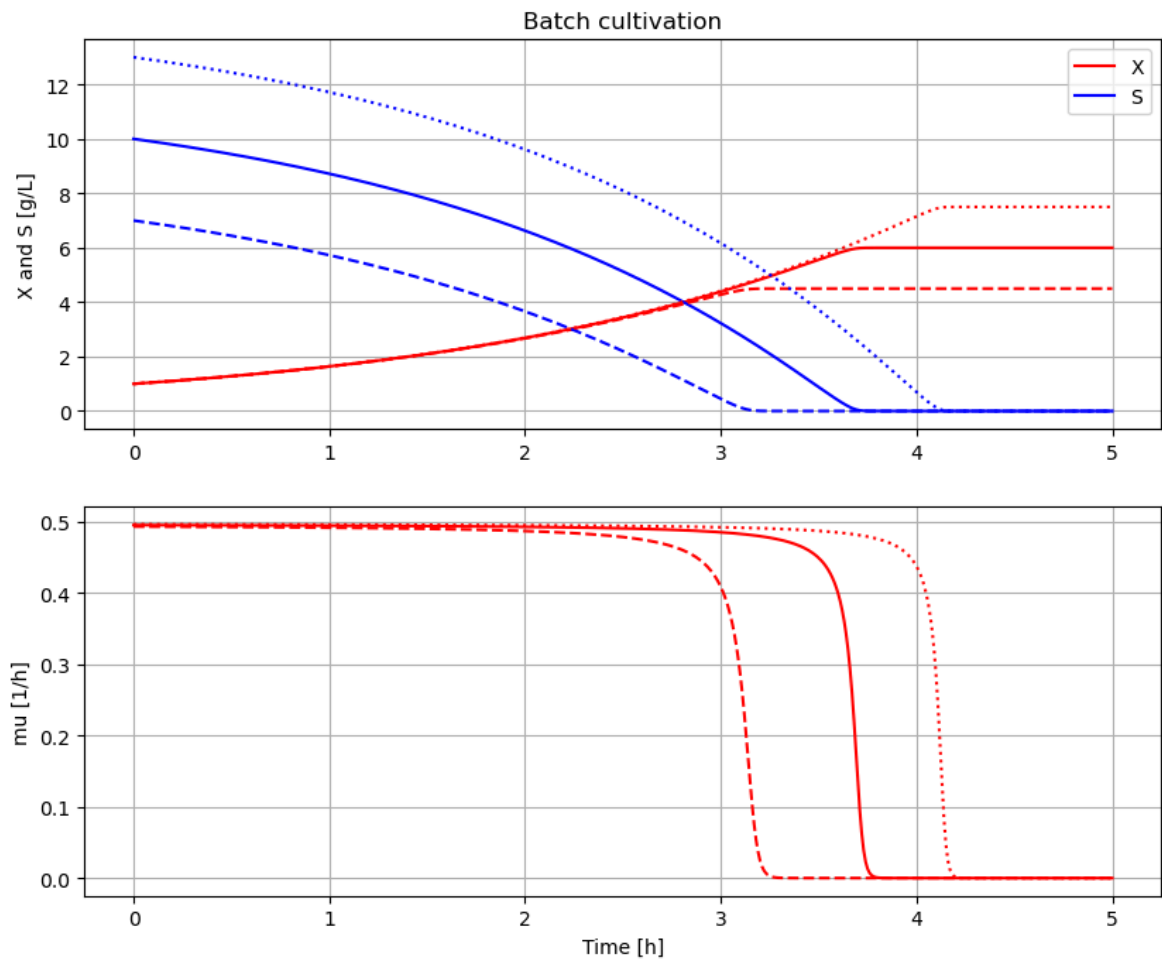
```
In [10]: # Simulation were initial value of biomass VX_0 is varied
newplot(plotType='TimeSeries')
for value in [1.0, 0.7, 1.3]: init(VX_start=value); simu(5)
```

```
# Restore default value of VX_0
init(VX_start=1.0)
```



```
In [11]: # Simulation were initial value of substrate VS_start is varied
newplot(plotType='TimeSeries')
for value in [10, 7, 13]: init(VS_start=value); simu(5)

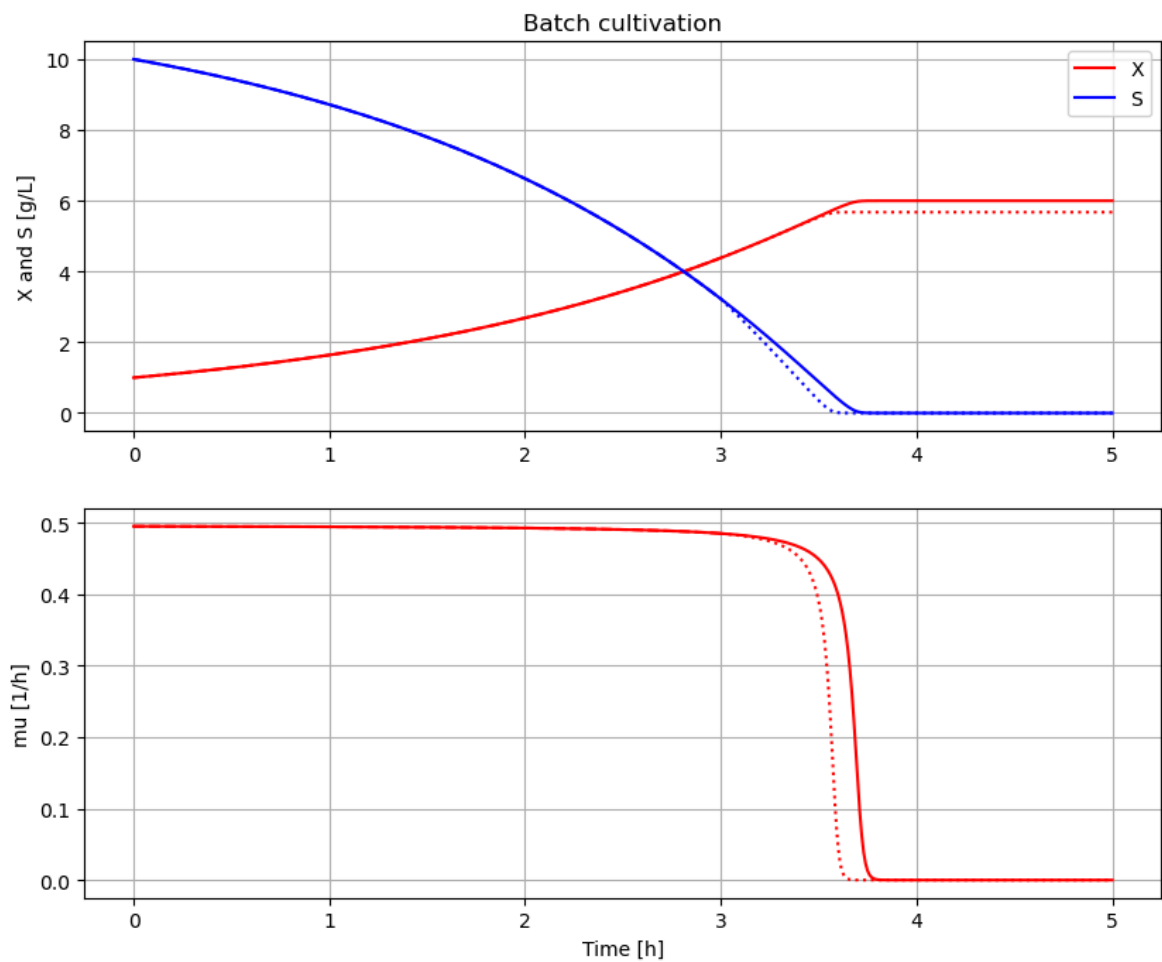
# Restore default value of VS_start
init(VS_start=10)
```



```
In [12]: # Simulation where metabolism is changed after 3 hours
newplot(plotType='TimeSeries')
simu(5)

simu(3)
par(Y=0.4, qSmax=1.0/(0.4/0.5)); simu(2, 'cont')

# Restore default value of Y and qSmax
par(Y=0.5, qSmax=1.0)
```



```
In [13]: disp('culture')
```

```
Y : 0.4
qSmax : 1.25
Ks : 0.1
```

```
In [14]: describe('mu')
```

```
Cell specific growth rate variable : 0.0 [ 1/h ]
```

```
In [15]: describe('parts')
```

```
['bioreactor', 'bioreactor.culture', 'MSL']
```

```
In [16]: describe('MSL')
```

```
MSL: none
```

```
In [17]: system_info()
```

System information

-OS: Windows
-Python: 3.12.11
-Scipy: 1.16.0
-PyFMI: 2.22.0
-FMU by: JModelica.org
-FMI: 2.0
-Type: FMUModelCS2
-Name: BPL.Examples_TEST2.Batch
-Generated: 2026-04-09T15:38:29
-MSL: 3.2.2 build 3
-Description: Bioprocess Library version 2.3.2
-Interaction: FMU_explore version 1.2.1

In [18]: `SDG('explan')`

"Soli Deo Gloria"

It is latin and means "To the honour of God".

The great composer Johan Sebastian Bach used to end his compositions with this small remark SDG.

And I like to do that too :).

In []: